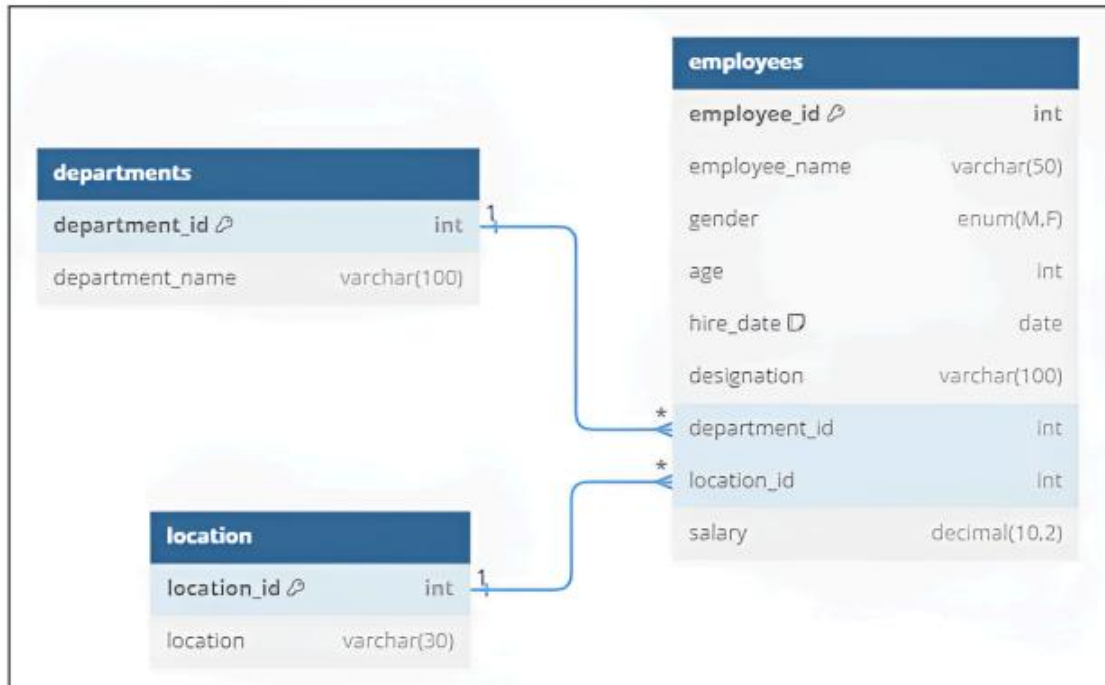


MySQL Assignment 1 – DDL Commands & Constraints

Employee Database schema:



DDL Commands :

Table Creation (CREATE): Write the SQL statements to create a database named “employee” and the following tables based on the provided schema:

The screenshot shows a MySQL IDE interface with the following components:

- Query 1:**

```
1 CREATE DATABASE employee;  
2  
3 USE employee;
```
- SQLAdditions:** Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
- Output:**

#	Time	Action	Message	Duration / Fetch
1	21:29:50	CREATE DATABASE employee	Error Code: 1007. Can't create database 'employee'; database exists	0.000 sec
2	21:31:10	DROP DATABASE IF EXISTS employee	3 row(s) affected	0.047 sec
3	21:31:29	CREATE DATABASE employee	1 row(s) affected	0.000 sec
4	21:31:29	USE employee	0 row(s) affected	0.016 sec

Departments

Query 1 x

SQLAdditions:.....

Limit to 1000 rows

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
1 CREATE TABLE departments (  
2   department_id INT,  
3   department_name VARCHAR(100)  
4 )
```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	21:29:50	CREATE DATABASE employee	Error Code: 1007. Can't create database 'employee'; database exists	0.000 sec
2	21:31:10	DROP DATABASE IF EXISTS employee	3 row(s) affected	0.047 sec
3	21:31:29	CREATE DATABASE employee	1 row(s) affected	0.000 sec
4	21:31:29	USE employee	0 row(s) affected	0.016 sec
5	21:32:28	CREATE DATABASE employee	Error Code: 1007. Can't create database 'employee'; database exists	0.016 sec
6	21:32:42	CREATE TABLE departments (department_id INT, department_name VARCHAR(100))	0 row(s) affected	0.015 sec

Location

Query 1 x

SQLAdditions:.....

Limit to 1000 rows

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
1 CREATE TABLE location (  
2   location_id INT,  
3   location VARCHAR(30)  
4 )
```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	21:31:10	DROP DATABASE IF EXISTS employee	3 row(s) affected	0.047 sec
3	21:31:29	CREATE DATABASE employee	1 row(s) affected	0.000 sec
4	21:31:29	USE employee	0 row(s) affected	0.016 sec
5	21:32:28	CREATE DATABASE employee	Error Code: 1007. Can't create database 'employee'; database exists	0.016 sec
6	21:32:42	CREATE TABLE departments (department_id INT, department_name VARCHAR(100))	0 row(s) affected	0.015 sec
7	21:33:13	CREATE TABLE location (location_id INT, location VARCHAR(30))	0 row(s) affected	0.015 sec

🕒 Employees

The screenshot shows a SQL IDE interface. The main editor displays the following SQL code:

```
1 CREATE TABLE employees (  
2     employee_id INT,  
3     employee_name VARCHAR(50),  
4     gender ENUM('M','F'),  
5     age INT,  
6     hire_date DATE,  
7     designation VARCHAR(100),  
8     department_id INT,  
9     location_id INT,  
10    salary DECIMAL(10,2)  
11 );
```

On the right side, there is a panel titled "SQLAdditions" with the text: "Automatic context help is d current care!". Below this, there are tabs for "Context Help" and "Snippets".

At the bottom, there is an "Output" panel with a dropdown menu set to "Action Output". It contains a table with the following data:

#	Time	Action	Message
✓ 3	21:31:29	CREATE DATABASE employee	1 row(s) affected
✓ 4	21:31:29	USE employee	0 row(s) affected
✗ 5	21:32:28	CREATE DATABASE employee	Error Code: 1007. Can't cr
✓ 6	21:32:42	CREATE TABLE departments (department_id INT, department_name VARCHAR(100))	0 row(s) affected
✓ 7	21:33:13	CREATE TABLE location (location_id INT, location VARCHAR(30))	0 row(s) affected
✓ 8	21:33:45	CREATE TABLE employees (employee_id INT, employee_name VARCHAR(50), gen...	0 row(s) affected

2. Table Alteration (ALTER): Consider the following scenarios and write the SQL

statements to alter the structure of the tables accordingly:

➊ Add a new column named "email" to the Employees table to store employee email addresses.

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window contains the following SQL statement:

```
1 ALTER TABLE employees
2 ADD email VARCHAR(255);
```

The 'Output' window displays the execution results:

#	Time	Action	Message
4	21:31:29	USE employee	0 row(s) affected
5	21:32:28	CREATE DATABASE employee	Error Code: 1007. Can't create database 'employee'; datab...
6	21:32:42	CREATE TABLE departments (department_id INT, department_name VARCHAR(100))	0 row(s) affected
7	21:33:13	CREATE TABLE location (location_id INT, location VARCHAR(30))	0 row(s) affected
8	21:33:45	CREATE TABLE employees (employee_id INT, employee_name VARCHAR(50), gen...	0 row(s) affected
9	21:49:11	ALTER TABLE employees ADD email VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

➋ Modify the data type of the "designation" column in the Employees table to support a wider range of values.

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window contains the following SQL statement:

```
1 ALTER TABLE employees
2 MODIFY designation VARCHAR(255);
```

The 'Output' window displays the execution results:

#	Time	Action	Message	Duration / Fetch
5	21:32:28	CREATE DATABASE employee	Error Code: 1007. Can't create database 'employee'; database exists	0.016 sec
6	21:32:42	CREATE TABLE departments (department_id INT, department_name VARCHAR(100))	0 row(s) affected	0.015 sec
7	21:33:13	CREATE TABLE location (location_id INT, location VARCHAR(30))	0 row(s) affected	0.015 sec
8	21:33:45	CREATE TABLE employees (employee_id INT, employee_name VARCHAR(50), gen...	0 row(s) affected	0.031 sec
9	21:49:11	ALTER TABLE employees ADD email VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
10	21:51:02	ALTER TABLE employees MODIFY designation VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec

🕒 Drop the “age” column from the Employees table.

The screenshot shows the SQL Developer interface with the following components:

- Query 1:** Contains two SQL statements:

```
1 ALTER TABLE employees
2 DROP COLUMN age;
```
- SQLAdditions:** A message stating: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."
- Output:** A table showing the execution results of the SQL statements.

#	Time	Action	Message	Duration / Fetch
6	21:32:42	CREATE TABLE departments (department_id INT, department_name VARCHAR(100))	0 row(s) affected	0.015 sec
7	21:33:13	CREATE TABLE location (location_id INT, location VARCHAR(30))	0 row(s) affected	0.015 sec
8	21:33:45	CREATE TABLE employees (employee_id INT, employee_name VARCHAR(50), gen...	0 row(s) affected	0.031 sec
9	21:49:11	ALTER TABLE employees ADD email VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
10	21:51:02	ALTER TABLE employees MODIFY designation VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
11	21:55:57	ALTER TABLE employees DROP COLUMN age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

🕒 Rename the “hire_date” column to “date_of_joining”.

The screenshot shows the SQL Developer interface with the following components:

- Query 1:** Contains two SQL statements:

```
1 ALTER TABLE employees
2 RENAME COLUMN hire_date TO date_of_joining;
```
- SQLAdditions:** A message stating: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."
- Output:** A table showing the execution results of the SQL statements.

#	Time	Action	Message	Duration / Fetch
7	21:33:13	CREATE TABLE location (location_id INT, location VARCHAR(30))	0 row(s) affected	0.015 sec
8	21:33:45	CREATE TABLE employees (employee_id INT, employee_name VARCHAR(50), gen...	0 row(s) affected	0.031 sec
9	21:49:11	ALTER TABLE employees ADD email VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
10	21:51:02	ALTER TABLE employees MODIFY designation VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
11	21:55:57	ALTER TABLE employees DROP COLUMN age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
12	21:56:35	ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec

3. Table Renaming (RENAME): Rewrite the SQL statements to rename the following tables:

☉ Rename the "Departments" table to "Departments_Info".

The screenshot shows the SQL Developer interface with a query window containing the statement: `RENAME TABLE departments TO Departments_Info;`. The output pane displays the execution results of this statement as part of a sequence of actions.

#	Time	Action	Message	Duration / Fetch
8	21:33:45	CREATE TABLE employees (employee_id INT, employee_name VARCHAR(50), gen...	0 row(s) affected	0.031 sec
9	21:49:11	ALTER TABLE employees ADD email VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
10	21:51:02	ALTER TABLE employees MODIFY designation VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
11	21:55:57	ALTER TABLE employees DROP COLUMN age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
12	21:56:35	ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
13	21:59:31	RENAME TABLE departments TO Departments_Info	0 row(s) affected	0.016 sec

☉ Rename the "Location" table to "Locations".

The screenshot shows the SQL Developer interface with a query window containing the statement: `ALTER TABLE departments TO Departments_Info; RENAME TABLE loca`. The output pane displays the execution results, showing a successful rename of the 'departments' table followed by a failed attempt to rename the 'locations' table.

#	Time	Action	Message	Duration / Fetch
9	21:49:11	ALTER TABLE employees ADD email VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
10	21:51:02	ALTER TABLE employees MODIFY designation VARCHAR(255)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
11	21:55:57	ALTER TABLE employees DROP COLUMN age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
12	21:56:35	ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
13	21:59:31	RENAME TABLE departments TO Departments_Info	0 row(s) affected	0.016 sec
14	22:00:42	RENAME TABLE departments TO Departments_Info	Error Code: 1050. Table 'departments_info' already exists	0.000 sec

4. Table Truncation (TRUNCATE): Write an SQL statement to truncate the Employees table.

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window contains the following SQL script:

```
1 USE employee;  
2  
3 TRUNCATE TABLE employees;
```

The 'Output' window displays the execution results of the script. The table below summarizes the actions performed:

#	Time	Action	Message	Duration / Fetch
✓ 12	21:56:35	ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
✓ 13	21:59:31	RENAME TABLE departments TO Departments_Info	0 row(s) affected	0.016 sec
✗ 14	22:00:42	RENAME TABLE departments TO Departments_Info	Error Code: 1050. Table 'departments_info' already exists	0.000 sec
✗ 15	22:03:59	RENAME TABLE departments TO Departments_Info	Error Code: 1050. Table 'departments_info' already exists	0.000 sec
✓ 16	22:07:06	USE employee	0 row(s) affected	0.000 sec
✓ 17	22:07:06	TRUNCATE TABLE employees	0 row(s) affected	0.032 sec

5. Database & Table Dropping (DROP): Write the SQL statements to drop the Employees table and then the “employee” database.

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window contains the following SQL script:

```
1 DROP TABLE employees;
```

The 'Output' window displays the execution results of the script. The table below summarizes the actions performed:

#	Time	Action	Message	Duration / Fetch
✓ 13	21:59:31	RENAME TABLE departments TO Departments_Info	0 row(s) affected	0.016 sec
✗ 14	22:00:42	RENAME TABLE departments TO Departments_Info	Error Code: 1050. Table 'departments_info' already exists	0.000 sec
✗ 15	22:03:59	RENAME TABLE departments TO Departments_Info	Error Code: 1050. Table 'departments_info' already exists	0.000 sec
✓ 16	22:07:06	USE employee	0 row(s) affected	0.000 sec
✓ 17	22:07:06	TRUNCATE TABLE employees	0 row(s) affected	0.032 sec
✓ 18	22:08:50	DROP TABLE employees	0 row(s) affected	0.016 sec

Query 1 x

SQLAdditions:.....

Limit to 1000 rows

1 DROP DATABASE employee;

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	22:00:42	RENAME TABLE departments TO Departments_Info	Error Code: 1050. Table 'departments_info' already exists	0.000 sec
15	22:03:59	RENAME TABLE departments TO Departments_Info	Error Code: 1050. Table 'departments_info' already exists	0.000 sec
16	22:07:06	USE employee	0 row(s) affected	0.000 sec
17	22:07:06	TRUNCATE TABLE employees	0 row(s) affected	0.032 sec
18	22:08:50	DROP TABLE employees	0 row(s) affected	0.016 sec
19	22:09:55	DROP DATABASE employee	2 row(s) affected	0.031 sec

Constraints :

1. Database Recreation:

🕒 Drop the 'employee' database if it exists and recreate it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed

Query 1 x

SQLAdditions:.....

Limit to 1000 rows

1 DROP DATABASE IF EXISTS employee;

2

3 CREATE DATABASE employee;

4

5 USE employee;

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
17	22:07:06	TRUNCATE TABLE employees	0 row(s) affected	0.032 sec
18	22:08:50	DROP TABLE employees	0 row(s) affected	0.016 sec
19	22:09:55	DROP DATABASE employee	2 row(s) affected	0.031 sec
20	22:45:56	DROP DATABASE IF EXISTS employee	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'employee'; database doesn't exist	0.015 sec
21	22:45:56	CREATE DATABASE employee	1 row(s) affected	0.016 sec
22	22:45:56	USE employee	0 row(s) affected	0.000 sec

2. Departments Table:

🕒 Ensure that the "department_id" uniquely identifies each department.

The screenshot shows the SQL Developer interface. The query editor contains the following SQL code:

```
1 CREATE TABLE departments (  
2   department_id INT PRIMARY KEY,  
3   department_name VARCHAR(100)  
4 );
```

The right pane displays a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom pane shows the "Output" window with the "Action Output" tab selected. It displays a table of execution results:

#	Time	Action	Message	Duration / Fetch
18	22:08:50	DROP TABLE employees	0 row(s) affected	0.016 sec
19	22:09:55	DROP DATABASE employee	2 row(s) affected	0.031 sec
20	22:45:56	DROP DATABASE IF EXISTS employee	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'employee'; database doesn't exist	0.015 sec
21	22:45:56	CREATE DATABASE employee	1 row(s) affected	0.016 sec
22	22:45:56	USE employee	0 row(s) affected	0.000 sec
23	22:47:29	CREATE TABLE departments (department_id INT PRIMARY KEY, department_name V...	0 row(s) affected	0.031 sec

🕒 Set up constraints on the "department_name" to avoid duplicate and null entries.

The screenshot shows the SQL Developer interface. The query editor contains the following SQL code:

```
1 CREATE TABLE departments (  
2   department_id INT PRIMARY KEY,  
3   department_name VARCHAR(100) NOT NULL UNIQUE  
4 );
```

The right pane displays a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom pane shows the "Output" window with the "Action Output" tab selected. It displays a table of execution results:

#	Time	Action	Message	Duration / Fetch
20	22:45:56	DROP DATABASE IF EXISTS employee	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'employee'; database doesn't exist	0.015 sec
21	22:45:56	CREATE DATABASE employee	1 row(s) affected	0.016 sec
22	22:45:56	USE employee	0 row(s) affected	0.000 sec
23	22:47:29	CREATE TABLE departments (department_id INT PRIMARY KEY, department_name V...	0 row(s) affected	0.031 sec

3. Location Table:

- 🕒 Establish a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially.
- 🕒 Implement constraints to prevent the insertion of null and duplicate locations.

The screenshot shows a SQL IDE interface. The top pane displays the SQL query for creating a table named 'location' with two columns: 'location_id' (INT, AUTO_INCREMENT, PRIMARY KEY) and 'location' (VARCHAR(30), NOT NULL, UNIQUE). The bottom pane shows the 'Output' window with a table of execution results.

#	Time	Action	Message	Duration / Fetch
21	22:45:56	CREATE DATABASE employee	1 row(s) affected	0.016 sec
22	22:45:56	USE employee	0 row(s) affected	0.000 sec
23	22:47:29	CREATE TABLE departments (department_id INT PRIMARY KEY, department_name V...	0 row(s) affected	0.031 sec
24	22:48:43	CREATE TABLE departments (department_id INT PRIMARY KEY, department_name V...	Error Code: 1050. Table 'departments' already exists	0.000 sec
25	22:49:30	CREATE TABLE departments (department_id INT PRIMARY KEY, department_name V...	Error Code: 1050. Table 'departments' already exists	0.015 sec
26	22:51:35	CREATE TABLE location (location_id INT AUTO_INCREMENT PRIMARY KEY, locatio...	0 row(s) affected	0.047 sec

4. Employees Table:

- 🕒 Guarantee that each employee has a distinct identifier.
- 🕒 Create a restriction to ensure that the employee's name is always provided.
- 🕒 Limit the acceptable values for the "gender" field to only 'M' or 'F'.
- 🕒 Enforce a condition to ensure that the employee's age is 18 or above.

⦿ Automatically assign the current date to the "hire_date" field if not specified.

⦿ Establish links between the "department_id" and "location_id" fields in the "employees" table and their respective tables.

Query 1

```
1 CREATE TABLE employees (  
2   employee_id INT PRIMARY KEY,  
3   employee_name VARCHAR(50) NOT NULL,  
4   gender ENUM('M', 'F'),  
5   age INT CHECK (age >= 18),  
6   hire_date DATE DEFAULT (CURRENT_DATE),  
7   designation VARCHAR(100),  
8   department_id INT,  
9   location_id INT,  
10  salary DECIMAL(10,2),  
11  
12  FOREIGN KEY (department_id)  
13    REFERENCES departments(department_id),  
14  
15  FOREIGN KEY (location_id)  
16    REFERENCES location(location_id)
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

#	Time	Action	Message	Duration / Fetch
23	22:47:29	CREATE TABLE departments (department_id INT PRIMARY KEY, department_name V...	0 row(s) affected	0.031 sec
24	22:48:43	CREATE TABLE departments (department_id INT PRIMARY KEY, department_name V...	Error Code: 1050. Table 'departments' already exists	0.000 sec
25	22:49:30	CREATE TABLE departments (department_id INT PRIMARY KEY, department_name V...	Error Code: 1050. Table 'departments' already exists	0.015 sec
26	22:51:35	CREATE TABLE location (location_id INT AUTO_INCREMENT PRIMARY KEY, locatio...	0 row(s) affected	0.047 sec
27	22:54:42	CREATE TABLE employees (employee_id INT PRIMARY KEY)	0 row(s) affected	0.031 sec
28	22:56:01	CREATE TABLE employees (employee_id INT PRIMARY KEY, employee_name VARC...	Error Code: 1050. Table 'employees' already exists	0.000 sec